National University of Colombia
Software Engineering II

Workshop 2

Design Artifacts and System Modeling

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1. System Domain

1.1 System Context

Purpose and problem

The system is a web platform that lets cyclists plan trips inside Bogotá considering factors that conventional navigation tools rarely prioritize: cycle-lane infrastructure, reported incidents, road condition and a safety level. Its goal is to help cyclists choose a route that is not only short or fast but also suitable for riding a bicycle.

The problem it addresses is that cyclists find it hard to compare routes using bicycle-specific criteria: a route may be shorter yet cross an area with many incidents or lack bicycle infrastructure. The literature cited in the project also notes that perceived insecurity discourages cycling, that near-misses and minor incidents are rarely reported to authorities, and that information on how dangerous each street is rarely available. Because the studies were done in Berlin and Zürich, the team assumes a similar dynamic in Bogotá, supported by a team member's experience as an active cyclist and by the official cycle-route data.

Scope

The system uses external map, geocoding and routing services. The project develops the cyclist-specific logic: it evaluates route alternatives by distance, time, cycle-lane use and safety; it stores community incident reports; and it builds a safety index from them to inform cyclists and recommend routes.

In scope	Out of scope
• Route planning and comparison in Bogotá only. • Community incident reporting, voting and flagging. • Safety index, unsafe zones and heat map. • Active navigation with alerts and re-routing. • User profile, history, reputation and administration.	• Building a proprietary cartographic system. • Competing with existing navigation platforms. • Cities other than Bogotá. • Routing outside bicycle profiles.

System boundary and external systems

External system	Role in the system
OpenStreetMap (OSM)	Base cartography displayed with Leaflet. Attribution must be visible.
Nominatim	Geocodes typed addresses. Limit: 1 request/s and an identifiable User-Agent.
openrouteservice (ORS)	Provides bicycle route alternatives and base travel times, subject to plan quotas.

External system	Role in the system
“Datos Abiertos Bogotá”	Official cycle-route layer (Secretaría Distrital de Movilidad, CC BY 4.0). Used instead of OSM alone because OSM cycle-lane mapping may be incomplete.
Device GPS	Gives current location for origin, reports and navigation, with the user’s permission.
E-mail service	Sends password-recovery links.

Technical constraints and initial conditions

- Technology: Python/Django backend (REST API, stateless), Vue.js frontend, Leaflet, PostgreSQL, OSM, Nominatim and ORS.
- Geography: only Bogotá.
- Data: at the start there are no reports, so the safety index begins with no history; test data may be seeded from Datos Abiertos Bogotá.
- Interface: Spanish, responsive from 360 px wide (used from a phone on the bicycle).
- Reliability: if ORS or Nominatim fail, the map and incidents remain available and a clear message is shown.
- Privacy: compliance with Law 1581 of 2012; location history is private.

1.2 Actor Characterization

Two human actors use the system. In addition, several external systems and one automatic process interact with it.

Human actors

Aspect	Cyclist	Administrator
Type	Primary actor, registered user.	Privileged actor, system supervisor.
Description	Person who rides in Bogotá and uses the system to plan safer routes and share road conditions.	Person who manages users and supervises the reports made by the community.
Goals	• Find and compare routes suited to cycling. • Know the risk of each route and area. • Alert others about hazards. • Stay safe during a ride.	• Keep the community safe and respectful. • Keep the quality of safety information high. • Monitor the platform’s overall state.

Aspect	Cyclist	Administrator
Main interactions	• Search routes, set preferences, compare alternatives. • View cycle-route, incident and unsafe-zone layers. • Create, edit, withdraw, confirm and flag reports. • Navigate with alerts and re-routing. • Manage profile, history and account.	• Search, filter, suspend, reactivate and terminate accounts. • Review, validate, edit and delete reports. • Consult the statistics dashboard.
Permissions	Everything after login except administration. Edits or withdraws only own reports.	Administration functions: user management, moderation and statistics.
Restrictions	Needs an active session; cannot flag or vote abusively; max. 5 reports per hour; Bogotá only.	Cannot reactivate a terminated account (permanent).
Related user stories	HU-C1 to HU-C23	HU-A1 to HU-A4

Secondary and system actors

Actor	Type	Interaction
Nominatim	External system	Receives addresses and returns coordinates.
openrouteservice	External system	Receives origin/destination and returns bicycle route alternatives.
Datos Abiertos Bogotá	External data source	Supplies the official cycle-route layer.
Scheduler (decay task)	Automatic process	Periodically reduces the weight of old reports and expires them.

1.3 Business Rules

Numeric values are initial reference values stored in a configuration file (not in the code) and may be adjusted after testing with real data.

Accounts and access

ID	Rule
BR-01	The e-mail used to register must be unique. Users consent to the privacy notice when registering.

ID	Rule
BR-02	All functions require an active session; without it only login and registration are visible.
BR-03	The system distinguishes the roles Cyclist and Administrator and restricts functions by role on every endpoint.
BR-04	After 5 failed login attempts the account is blocked for 15 minutes.
BR-05	A session or token expires after 24 hours.
BR-06	The password-recovery link is valid for 1 hour and can be used once.
BR-07	Suspended accounts can be reactivated; terminated accounts cannot (the termination is permanent).
BR-08	When a user deletes their account, personal data and history are erased, and the reports they made no longer show identifying data.

Geography and routes

ID	Rule
BR-09	Origin, destination and report locations must be inside Bogotá; otherwise the system informs the user and does not calculate or create anything.
BR-10	Each search requests 3 route alternatives from ORS using the bicycle profile.
BR-11	Alternatives are ordered by the user's priority (speed, cycle lanes or safety) and exclude high-speed or unpaved roads if requested.
BR-12	Travel time is recalculated with a reference speed of 15 km/h, reduced on slopes greater than 5 %.
BR-13	The comparative panel shows distance, estimated time, safety rating and the percentage of the route on official cycle routes.

Safety index

ID	Rule
BR-14	A segment without incidents has a base score of 100.
BR-15	Each active report penalizes segments within a 100 m radius according to severity (low = 1, medium = 3, high = 5) and type (theft/harassment = 5, accident = 4, obstruction = 3, pothole/flooding = 2, roadwork = 1).
BR-16	A segment on an official cycle route receives a bonus of +15 points.
BR-17	A report with at least 3 confirmations receives more weight.
BR-18	The report's weight decays with a half-life of 14 days; a report with no confirmations expires after 60 days.

ID	Rule
BR-19	Each route receives a score from 0 to 100: green ≥ 70, yellow 40–69, red < 40. Color is always accompanied by text or an icon.
BR-20	An area with an index below 40 is an unsafe zone.

Community reports

ID	Rule
BR-21	A report requires type (theft, pothole, flooding, roadwork, harassment, accident, obstruction, other), severity (low, medium, high), description and location.
BR-22	A user can create at most 5 reports per hour.
BR-23	If a report of the same type exists within 50 m and the last 24 h, the user is warned and offered to confirm it instead.
BR-24	Each user has one vote per report (still active or solved); 3 “solved” votes mark the report as resolved.
BR-25	A report has one state: active, resolved, expired or moderated.
BR-26	Only the author can edit or withdraw a report; withdrawing it removes its effect on the index.
BR-27	A user can flag a report once, and cannot flag their own; flagged reports enter the administrator’s review queue.
BR-28	Public reports never reveal the identity of their author.
BR-29	Points and reputation depend on the quantity and veracity of a user’s reports.

Navigation

ID	Rule
BR-30	A ride needs a chosen route and location permission; it can be started, paused and ended.
BR-31	The user is alerted of incidents within 300 m; alerts stop while paused.
BR-32	A deviation of more than 50 m from the route triggers a new safe route from the current location.
BR-33	If GPS signal is lost, the system tells the user instead of showing an outdated position.

Administration

ID	Rule
BR-34	Only administrators can manage accounts (search, suspend, reactivate, terminate).
BR-35	Only administrators can validate, edit or delete reports, including those flagged.

External services and compliance

ID	Rule
BR-36	OSM attribution is visible; Nominatim usage is limited to 1 request per second with an identifiable User-Agent; ORS quotas are respected.
BR-37	The cycle-route layer carries attribution to the Secretaría Distrital de Movilidad (CC BY 4.0).
BR-38	API keys and credentials are kept in environment variables, never in the repository or the frontend.

2. CRC Cards

This section identifies the main classes of the system. Each card lists the class name, its responsibilities (what it knows and does) and its collaborators (the classes it works with to fulfil them). Classes were derived from the requirements, user stories and business rules.

2.1 Class Overview

#	Class	Description
1	User	Abstract base class for every person who holds an account in the system.
2	Cyclist	Primary actor. Plans routes, reports incidents and participates in the community.
3	Administrator	Privileged actor who supervises users and community content.
4	UserProfile	Personal data and optional cycling attributes of a user.
5	TravelPreferences	Route priorities and restrictions chosen by a cyclist.
6	SearchHistory	Record of recent searches and routes used by a cyclist.
7	ReputationAccount	Points and reputation level earned through community reports.
8	AuthenticationService	Controller for registration, login, logout and password recovery.
9	RoutePlanner	Controller that orchestrates route search, evaluation and ordering.
10	RouteAlternative	One candidate route between origin and destination.
11	RouteSegment	Stretch of street that is the unit of safety evaluation.
12	RoutingProvider	Adapter to openrouteservice (bicycle profile).
13	GeocodingService	Adapter to Nominatim for addresses and coordinates.
14	CycleRoute	Official cycle-route network from Datos Abiertos Bogotá.
15	SafetyIndexCalculator	Isolated module with the safety-index logic, independent of the routing provider.
16	IncidentReport	Community report of a hazard at a location.
17	ReportVote	A user’s vote on the validity of a report.
18	ReportFlag	Complaint that a report is false or inappropriate.
19	AbuseControl	Rate limiting and duplicate-detection rules.
20	DecayScheduler	Scheduled task that ages reports.
21	NavigationSession	Active ride followed by a cyclist.

#	Class	Description
22	AlertService	Real-time warnings during navigation.
23	MapLayerService	Provides the optional layers over the OSM base map.
24	ModerationService	Administrator tools for user and content moderation.
25	StatisticsDashboard	Global metrics for administrators.
26	SystemConfiguration	Configuration file holding business-rule parameters.

2.2 CRC Cards

CRC-01 | Class: User
Abstract base class for every person who holds an account in the system.

Responsibilities	Collaborators
• Store identity data: name, unique e-mail, password hash, account status (active, suspended, terminated). • Authenticate with credentials and keep a session or token (24 h expiry). • Expose the role (Cyclist or Administrator) used for access control. • Log out and invalidate the session.	• AuthenticationService • Cyclist • Administrator • UserProfile

CRC-02 | Class: Cyclist
Primary actor. Plans routes, reports incidents and participates in the community.

Responsibilities	Collaborators
• Request routes between an origin and a destination inside Bogotá. • Create, edit and withdraw own incident reports. • Vote on the validity of reports and flag false or inappropriate ones. • Start, pause and end a ride (navigation). • Manage own profile, preferences, history and account deletion.	• User • RoutePlanner • IncidentReport • ReportVote • ReportFlag • NavigationSession • UserProfile • TravelPreferences • SearchHistory • ReputationAccount

CRC-03 | Class: Administrator *Privileged actor who supervises users and community content.*	
Responsibilities	**Collaborators**
• Search, filter, suspend, reactivate and terminate user accounts. • Review, validate, edit and delete reports, including flagged ones. • Consult the statistics dashboard.	• User • ModerationService • StatisticsDashboard • IncidentReport • ReportFlag

CRC-04 | Class: UserProfile *Personal data and optional cycling attributes of a user.*	
Responsibilities	**Collaborators**
• Hold optional data: bicycle type and fitness level. • Allow the cyclist to edit those fields. • Be removed completely when the account is deleted (Law 1581 of 2012).	• Cyclist • TravelPreferences

CRC-05 | Class: TravelPreferences *Route priorities and restrictions chosen by a cyclist.*	
Responsibilities	**Collaborators**
• Store priority (speed, cycle lanes or safety) and restrictions (avoid high-speed roads, avoid unpaved streets). • Save default preferences for the user. • Supply the ordering criterion applied to route alternatives.	• Cyclist • UserProfile • RoutePlanner • RouteAlternative

CRC-06 | Class: SearchHistory *Record of recent searches and routes used by a cyclist.*	
Responsibilities	**Collaborators**

• Save recent searches and selected routes. • List them in chronological order as quick access. • Keep location history private to its owner. • Be deleted with the account.	• Cyclist • RoutePlanner • RouteAlternative

CRC-07 | Class: ReputationAccount *Points and reputation level earned through community reports.*	
Responsibilities	**Collaborators**
• Accumulate points when reports are verified as true. • Compute and show the reputation level. • Reflect the quantity and veracity of a user's contributions.	• Cyclist • IncidentReport • ReportVote • ModerationService

CRC-08 | Class: AuthenticationService *Controller for registration, login, logout and password recovery.*	
Responsibilities	**Collaborators**
• Register users and enforce unique e-mail and consent to the privacy notice. • Hash passwords and validate credentials. • Issue and expire sessions or tokens; verify the role on every request. • Lock the account for 15 min after 5 failed logins. • Send a one-hour, single-use password recovery link.	• User • SystemConfiguration • AbuseControl • Administrator • Cyclist

CRC-09 | Class: RoutePlanner *Controller that orchestrates route search, evaluation and ordering.*	
Responsibilities	**Collaborators**

• Validate that origin and destination are inside Bogotá. • Request 3 bike alternatives through the routing provider. • Ask the safety module to score each alternative and compute its cycle-lane percentage. • Order alternatives according to the preferences and build the comparison data. • Show a clear message if external services fail.	• GeocodingService • RoutingProvider • RouteAlternative • SafetyIndexCalculator • CycleRoute • TravelPreferences • SearchHistory

CRC-10 | Class: RouteAlternative
One candidate route between origin and destination.

Responsibilities	Collaborators
• Hold geometry, distance, adjusted time, % on official cycle routes, safety score and color. • Be ordered and compared against other alternatives. • Provide the data shown in the comparative panel. • Adjust time using 15 km/h and a slope reduction above 5 %.	• RoutePlanner • RouteSegment • SafetyIndexCalculator • NavigationSession

CRC-11 | Class: RouteSegment
Stretch of street that is the unit of safety evaluation.

Responsibilities	Collaborators
• Know its geometry and whether it lies on an official cycle route. • Provide the incidents within its influence radius. • Keep its safety index (base 100).	• RouteAlternative • SafetyIndexCalculator • IncidentReport • CycleRoute

CRC-12 | Class: RoutingProvider
Adapter to openrouteservice (bicycle profile).

Responsibilities	Collaborators

• Request alternative routes and base times from ORS. • Translate the provider response into internal route objects. • Respect plan quotas and report failures without breaking the map. • Isolate the provider so the safety logic remains independent.	• RoutePlanner • RouteAlternative • SystemConfiguration

CRC-13 | Class: GeocodingService *Adapter to Nominatim for addresses and coordinates.*	
Responsibilities	**Collaborators**
• Convert a typed address into coordinates (and map clicks into points). • Reject points outside Bogotá. • Respect the limit of 1 request per second with an identifiable User-Agent. • Cache repeated geocodings.	• RoutePlanner • IncidentReport • SystemConfiguration

CRC-14 | Class: CycleRoute *Official cycle-route network from Datos Abiertos Bogotá.*	
Responsibilities	**Collaborators**
• Store the official cycle-route geometries (CC BY 4.0, attribution to the Secretaría Distrital de Movilidad). • Tell whether a segment lies on a cycle route. • Provide the layer displayed on the map.	• RouteSegment • RoutePlanner • MapLayerService • SafetyIndexCalculator

CRC-15 | Class: SafetyIndexCalculator *Isolated module with the safety-index logic, independent of the routing provider.*	
Responsibilities	**Collaborators**

• Compute each segment's index: base 100, minus penalties by severity, type, age and confirmations, plus +15 for cycle routes. • Aggregate segments into a 0–100 route score. • Assign color: green ≥ 70, yellow 40–69, red < 40. • Identify unsafe zones (index < 40). • Read all parameters from configuration.	• RouteSegment • RouteAlternative • IncidentReport • CycleRoute • SystemConfiguration • MapLayerService

CRC-16 | Class: IncidentReport *Community report of a hazard at a location.*	
Responsibilities	**Collaborators**
• Store type, severity, description, location, author, creation date and state. • Validate required data and that the location is in Bogotá. • Manage states: active, resolved, expired, moderated. • Be edited or withdrawn only by its author. • Hide the author's identity in public views. • Count confirmations and "solved" votes.	• Cyclist • ReportVote • ReportFlag • AbuseControl • SafetyIndexCalculator • GeocodingService • DecayScheduler • ModerationService

CRC-17 | Class: ReportVote *A user's vote on the validity of a report.*	
Responsibilities	**Collaborators**
• Record whether the incident is still active or solved. • Guarantee one vote per user and report. • Trigger "resolved" after 3 solved votes and extra weight after 3 confirmations.	• Cyclist • IncidentReport • ReputationAccount

CRC-18 | Class: ReportFlag *Complaint that a report is false or inappropriate.*	
Responsibilities	**Collaborators**

• Register the reason and the reporting user. • Prevent duplicate flags by the same user and flags on own reports. • Place the report in the administrator's review queue.	• Cyclist • IncidentReport • ModerationService

CRC-19 | Class: AbuseControl *Rate limiting and duplicate-detection rules.*	
Responsibilities	**Collaborators**
• Limit report creation to 5 per user per hour and tell when the user may retry. • Warn about a similar report (same type, < 50 m, < 24 h) and offer to confirm it. • Limit login attempts and apply temporary lockout.	• IncidentReport • AuthenticationService • Cyclist • SystemConfiguration

CRC-20 | Class: DecayScheduler *Scheduled task that ages reports.*	
Responsibilities	**Collaborators**
• Reduce report weight with a 14-day half-life. • Expire reports with no confirmations after 60 days. • Trigger index recalculation for affected segments.	• IncidentReport • SafetyIndexCalculator • SystemConfiguration

CRC-21 | Class: NavigationSession *Active ride followed by a cyclist.*	
Responsibilities	**Collaborators**
• Start, pause, resume and end a ride with real-time position. • Detect deviation (> 50 m) and request a new safe route. • Detect loss of GPS signal and inform the user. • Request alerts for incidents within 300 m.	• Cyclist • RouteAlternative • AlertService • RoutePlanner

CRC-22 | Class: AlertService *Real-time warnings during navigation.*	
Responsibilities	**Collaborators**
• Find active incidents near the current position (300 m). • Show on-screen alerts with text or icon, not only color. • Stop alerts while the ride is paused.	• NavigationSession • IncidentReport

CRC-23 | Class: MapLayerService *Provides the optional layers over the OSM base map.*	
Responsibilities	**Collaborators**
• Serve the incident layer with filters by type and severity and click-through detail. • Serve the official cycle-route layer. • Serve the heat map and unsafe-zone layer. • Show OSM and data attribution.	• IncidentReport • CycleRoute • SafetyIndexCalculator

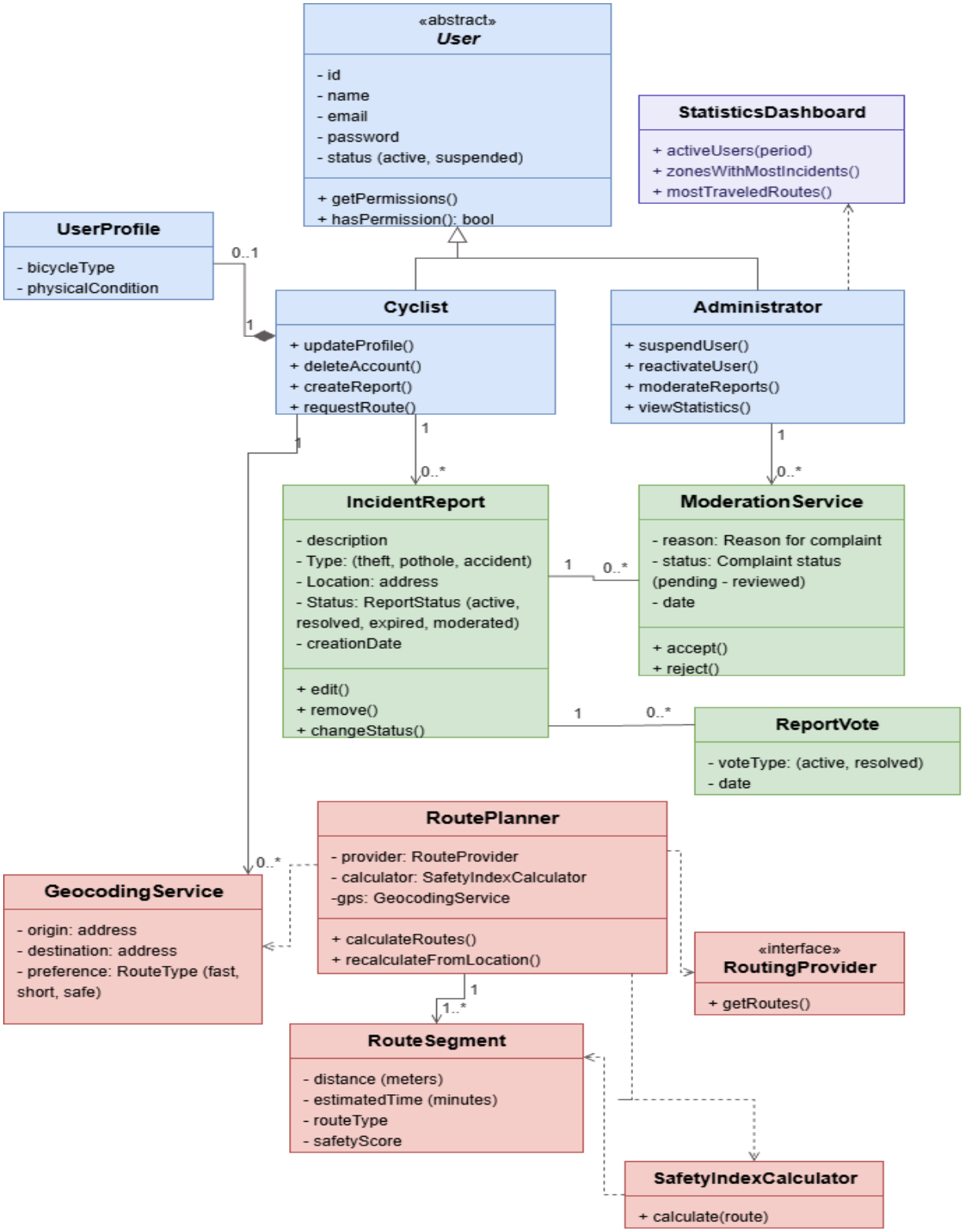
CRC-24 | Class: ModerationService *Administrator tools for user and content moderation.*	
Responsibilities	**Collaborators**
• Offer the moderation queue of flagged and doubtful reports. • Validate, edit or delete reports. • Suspend, reactivate or terminate accounts (termination is permanent).	• Administrator • IncidentReport • ReportFlag • User • ReputationAccount

CRC-25 | Class: StatisticsDashboard *Global metrics for administrators.*	
Responsibilities	**Collaborators**

• Show active users. • Show zones with most incidents. • Show which route type users choose (routing effectiveness).	• Administrator • IncidentReport • User • RoutePlanner

CRC-26 | Class: SystemConfiguration *Configuration file holding business-rule parameters.*	
Responsibilities	**Collaborators**
• Store thresholds, weights, radii, limits and times (see Section 1.3). • Provide values to other modules without hard-coding them. • Keep API keys and credentials in environment variables only.	• SafetyIndexCalculator • DecayScheduler • AbuseControl • AuthenticationService • RoutingProvider • GeocodingService

3. UML class diagram



4. Relational Database Model

4.1 Design decisions

The relational schema is derived from the classes in Section 2 and the class diagram in Section 3, and it applies the business rules in Section 1.3. It targets PostgreSQL, with the PostGIS extension for geometry columns, consistent with the technical constraints of the project.

- Cyclist and Administrator share a single users table, differentiated by the role column (single-table inheritance of the abstract User class).
- SystemConfiguration is a configuration file (BR-38), so it has no table. AlertService, AbuseControl and MapLayerService are logic without data of their own: the report rate limit (BR-22) is computed from incident_reports, and the login lockout (BR-04) uses columns of users.
- When an account is deleted (BR-08), personal data and history are removed through ON DELETE CASCADE, while incident_reports.author_id uses ON DELETE SET NULL so that reports remain without identifying data.
- Confirmation count, solved count and current weight are stored in incident_reports so that the DecayScheduler and the SafetyIndexCalculator can read them efficiently.
- Route alternatives and their segments are persisted to feed the search history, the navigation sessions and the statistics dashboard.

4.2 Table definitions

Each table is presented as a card with its columns, data types, primary keys (PK), foreign keys (FK) and constraints. The class each table implements is shown in parentheses.

DB-01 | Table: users
Stores every account (User, Cyclist and Administrator).

Column	Type	Constraints
user_id	BIGINT	PK
full_name	VARCHAR(100)	NOT NULL
email	VARCHAR(254)	NOT NULL, UNIQUE (BR-01)
password_hash	VARCHAR(255)	NOT NULL
role	VARCHAR(15)	NOT NULL, CHECK (cyclist, administrator) (BR-03)
status	VARCHAR(12)	NOT NULL, CHECK (active, suspended, terminated) (BR-07)
privacy_consent_at	TIMESTAMP	NOT NULL (BR-01)
failed_login_attempts	SMALLINT	NOT NULL, DEFAULT 0 (BR-04)
locked_until	TIMESTAMP	NULL (BR-04)
created_at	TIMESTAMP	NOT NULL, DEFAULT now()

DB-02 | Table: password_reset_tokens
Single-use recovery links sent by e-mail.

Column	Type	Constraints
token_id	BIGINT	PK
user_id	BIGINT	FK → users, ON DELETE CASCADE
token_hash	VARCHAR(255)	NOT NULL, UNIQUE
expires_at	TIMESTAMP	NOT NULL (creation time + 1 h) (BR-06)
used_at	TIMESTAMP	NULL (single use) (BR-06)

DB-03 | Table: user_profiles
Optional cycling attributes of a user (UserProfile).

Column	Type	Constraints
user_id	BIGINT	PK, FK → users, ON DELETE CASCADE
bicycle_type	VARCHAR(30)	NULL
fitness_level	VARCHAR(20)	NULL

DB-04 | Table: travel_preferences
Default route priorities and restrictions (TravelPreferences).

Column	Type	Constraints
user_id	BIGINT	PK, FK → users, ON DELETE CASCADE
priority	VARCHAR(12)	NOT NULL, CHECK (speed, cycle_lanes, safety) (BR-11)
avoid_high_speed_roads	BOOLEAN	NOT NULL, DEFAULT false
avoid_unpaved_streets	BOOLEAN	NOT NULL, DEFAULT false

DB-05 | Table: reputation_accounts
Points and reputation level of a user (ReputationAccount).

Column	Type	Constraints
user_id	BIGINT	PK, FK → users, ON DELETE CASCADE
points	INTEGER	NOT NULL, DEFAULT 0, CHECK (points >= 0) (BR-29)
level	VARCHAR(20)	NOT NULL, DEFAULT 'newcomer'

DB-06 | Table: search_history
Recent route searches, private to their owner (SearchHistory).

Column	Type	Constraints
search_id	BIGINT	PK
user_id	BIGINT	FK → users, ON DELETE CASCADE (BR-08)
origin_text	VARCHAR(255)	NOT NULL
origin_point	GEOMETRY(Point, 4326)	NOT NULL, inside Bogotá (BR-09)
destination_text	VARCHAR(255)	NOT NULL
destination_point	GEOMETRY(Point, 4326)	NOT NULL, inside Bogotá (BR-09)
priority	VARCHAR(12)	NOT NULL
searched_at	TIMESTAMP	NOT NULL, DEFAULT now()

DB-07 | Table: route_alternatives
Candidate routes produced by a search (RouteAlternative).

Column	Type	Constraints
alternative_id	BIGINT	PK
search_id	BIGINT	FK → search_history, ON DELETE CASCADE
rank_position	SMALLINT	NOT NULL, CHECK (1 to 3) (BR-10, BR-11)
geometry	GEOMETRY(LineString, 4326)	NOT NULL
distance_m	NUMERIC(10,2)	NOT NULL
adjusted_time_min	NUMERIC(8,2)	NOT NULL (BR-12)
cycle_route_pct	NUMERIC(5,2)	NOT NULL, CHECK (0 to 100) (BR-13)
safety_score	NUMERIC(5,2)	NOT NULL, CHECK (0 to 100) (BR-19)
safety_color	VARCHAR(6)	NOT NULL, CHECK (green, yellow, red) (BR-19)
was_selected	BOOLEAN	NOT NULL, DEFAULT false (used by the statistics dashboard)

DB-08 | Table: road_segments
Street stretches that are the unit of safety evaluation (RouteSegment).

Column	Type	Constraints
segment_id	BIGINT	PK
osm_way_id	BIGINT	NULL

geometry	GEOMETRY(LineString, 4326)	NOT NULL
cycle_route_id	BIGINT	FK → cycle_routes, NULL, ON DELETE SET NULL (BR-16)
safety_index	NUMERIC(5,2)	NOT NULL, DEFAULT 100 (BR-14)
last_calculated_at	TIMESTAMP	NULL

DB-09 | Table: alternative_segments
Junction table between route alternatives and road segments.

Column	Type	Constraints
alternative_id	BIGINT	PK (composite), FK → route_alternatives, ON DELETE CASCADE
segment_id	BIGINT	PK (composite), FK → road_segments
segment_order	SMALLINT	NOT NULL

DB-10 | Table: cycle_routes
Official cycle-route network from Datos Abiertos Bogotá (CycleRoute).

Column	Type	Constraints
cycle_route_id	BIGINT	PK
name	VARCHAR(150)	NULL
geometry	GEOMETRY(LineString, 4326)	NOT NULL
source_attribution	VARCHAR(255)	NOT NULL (BR-37)

DB-11 | Table: incident_reports
Community reports of hazards at a location (IncidentReport).

Column	Type	Constraints
report_id	BIGINT	PK
author_id	BIGINT	FK → users, NULL, ON DELETE SET NULL (BR-08, BR-28)
type	VARCHAR(15)	NOT NULL, CHECK (theft, pothole, flooding, roadwork, harassment, accident, obstruction, other) (BR-21)
severity	VARCHAR(6)	NOT NULL, CHECK (low, medium, high) (BR-21)
description	TEXT	NOT NULL

location	GEOMETRY(Point, 4326)	NOT NULL, inside Bogotá (BR-09)
address	VARCHAR(255)	NULL
state	VARCHAR(10)	NOT NULL, CHECK (active, resolved, expired, moderated) (BR-25)
confirmation_count	INTEGER	NOT NULL, DEFAULT 0 (BR-17)
solved_count	INTEGER	NOT NULL, DEFAULT 0 (BR-24)
current_weight	NUMERIC(6,4)	NOT NULL, DEFAULT 1 (BR-18)
created_at	TIMESTAMP	NOT NULL, DEFAULT now()
updated_at	TIMESTAMP	NOT NULL

DB-12 | Table: report_votes
One vote per user and report (ReportVote).

Column	Type	Constraints
vote_id	BIGINT	PK
report_id	BIGINT	FK → incident_reports, ON DELETE CASCADE
user_id	BIGINT	FK → users, ON DELETE CASCADE
vote_type	VARCHAR(8)	NOT NULL, CHECK (active, solved)
created_at	TIMESTAMP	NOT NULL, DEFAULT now()
Table constraint: UNIQUE (report_id, user_id) (BR-24)		

DB-13 | Table: report_flags
Complaints that a report is false or inappropriate (ReportFlag).

Column	Type	Constraints
flag_id	BIGINT	PK
report_id	BIGINT	FK → incident_reports, ON DELETE CASCADE
flagged_by	BIGINT	FK → users, ON DELETE CASCADE
reason	VARCHAR(255)	NOT NULL
status	VARCHAR(10)	NOT NULL, DEFAULT 'pending', CHECK (pending, accepted, rejected)
reviewed_by	BIGINT	FK → users (administrator), NULL
created_at	TIMESTAMP	NOT NULL, DEFAULT now()
reviewed_at	TIMESTAMP	NULL
Table constraint: UNIQUE (report_id, flagged_by) (BR-27)		

DB-14 | Table: moderation_actions
Audit log of administrator actions (ModerationService).

Column	Type	Constraints
action_id	BIGINT	PK
admin_id	BIGINT	FK → users (administrator), NOT NULL
target_user_id	BIGINT	FK → users, NULL
target_report_id	BIGINT	FK → incident_reports, NULL
action_type	VARCHAR(20)	NOT NULL, CHECK (suspend, reactivate, terminate, validate, edit, delete) (BR-34, BR-35)
notes	TEXT	NULL
created_at	TIMESTAMP	NOT NULL, DEFAULT now()
Table constraint: CHECK (target_user_id IS NOT NULL OR target_report_id IS NOT NULL)		

DB-15 | Table: navigation_sessions
Rides followed by a cyclist (NavigationSession).

Column	Type	Constraints
session_id	BIGINT	PK
user_id	BIGINT	FK → users, ON DELETE CASCADE
alternative_id	BIGINT	FK → route_alternatives, NOT NULL
status	VARCHAR(8)	NOT NULL, CHECK (active, paused, ended) (BR-30)
started_at	TIMESTAMP	NOT NULL
ended_at	TIMESTAMP	NULL

DB-16 | Table: geocoding_cache
Cache of repeated geocoding requests (GeocodingService).

Column	Type	Constraints
query_text	VARCHAR(255)	PK
location	GEOMETRY(Point, 4326)	NOT NULL
cached_at	TIMESTAMP	NOT NULL, DEFAULT now()

4.3 Relationships

The following table summarizes the relationships between tables, including their cardinality.

Parent table	Child table	Cardinality	Foreign key
users	user_profiles	1 : 0..1	user_profiles.user_id
users	travel_preferences	1 : 0..1	travel_preferences.user_id
users	reputation_accounts	1 : 0..1	reputation_accounts.user_id
users	password_reset_tokens	1 : 0..*	password_reset_tokens.user_id
users	search_history	1 : 0..*	search_history.user_id
users	incident_reports	1 : 0..*	incident_reports.author_id
users	report_votes	1 : 0..*	report_votes.user_id
users	report_flags	1 : 0..*	report_flags.flagged_by
users (administrator)	moderation_actions	1 : 0..*	moderation_actions.admin_id
users	navigation_sessions	1 : 0..*	navigation_sessions.user_id
incident_reports	report_votes	1 : 0..*	report_votes.report_id
incident_reports	report_flags	1 : 0..*	report_flags.report_id
incident_reports	moderation_actions	1 : 0..*	moderation_actions.target_report_id
search_history	route_alternatives	1 : 1..3	route_alternatives.search_id
route_alternatives	alternative_segments	1 : 1..*	alternative_segments.alternative_id
road_segments	alternative_segments	1 : 0..*	alternative_segments.segment_id
cycle_routes	road_segments	0..1 : 0..*	road_segments.cycle_route_id
route_alternatives	navigation_sessions	1 : 0..*	navigation_sessions.alternative_id

4.4 ER diagram 5.

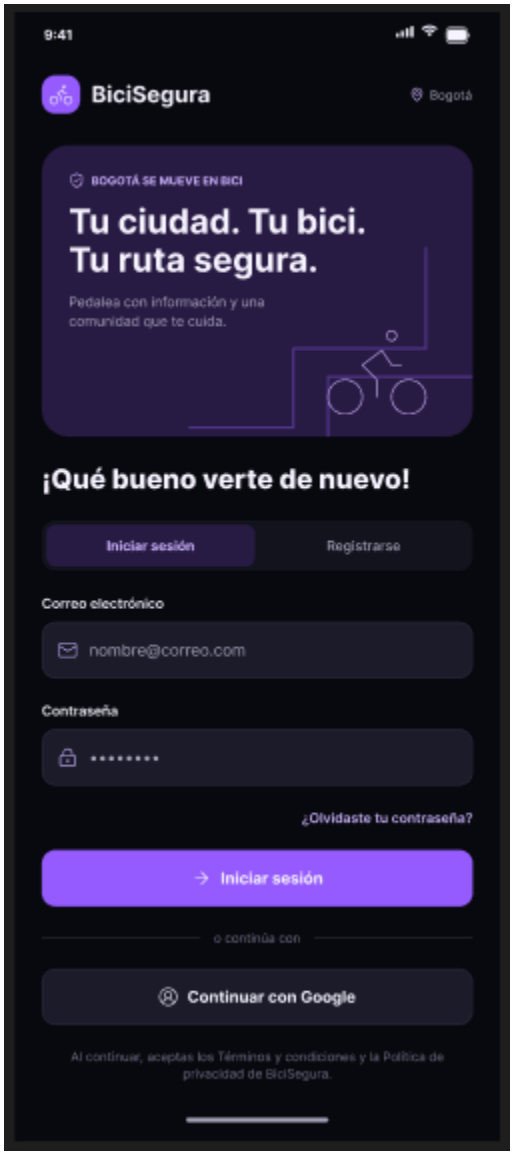
The entity-relationship diagram, which shows the tables, their keys and the relationships described in Section 4.3, is included in the Annex due to its size.

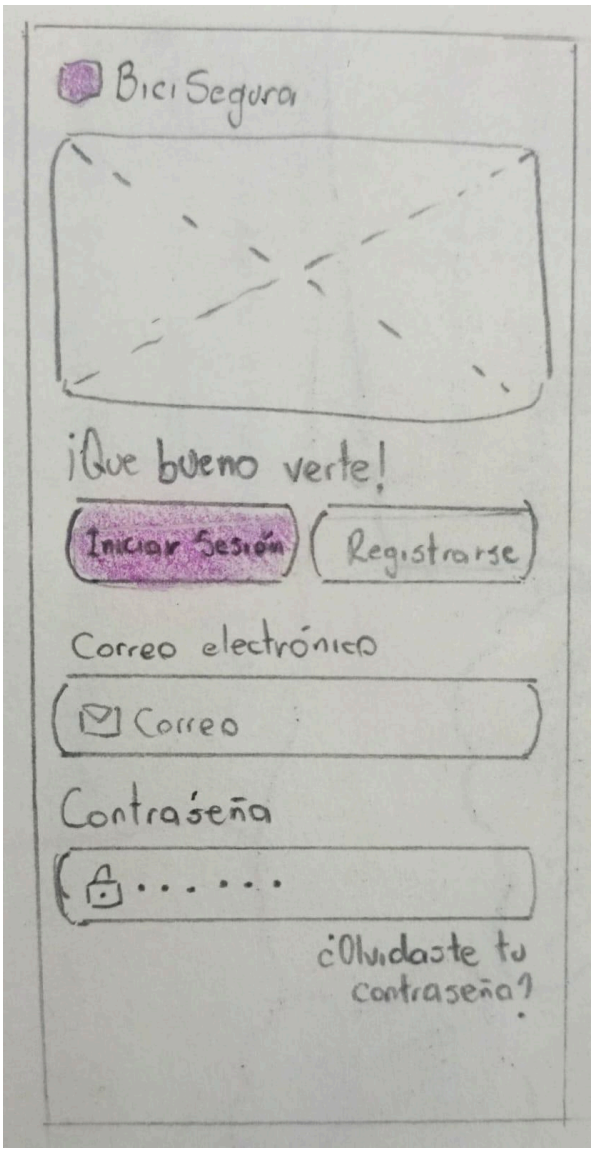
5. Mockups

The interface design for BiciSegura was developed in two stages. First, five low-fidelity mockups were sketched by hand to define the basic structure and flow of the screens. Then, using Figma, eleven high-fidelity mockups were created covering both the cyclist and the administrator flows. The complete set of high-fidelity designs is available at the following link: https://www.figma.com/design/aFUXesQLkr5MsEGJWTHqhF/Sin-t%C3%ADtulo?node-id=0-1&t=Fuo3DzJCKRKNZ7EF-1. The seven most important screens are described below, along with the purpose of each one.

Login & Register

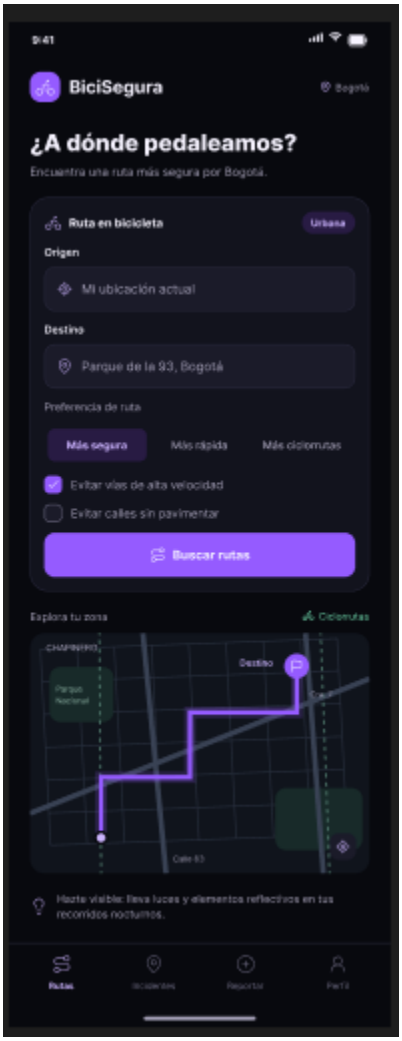
This is the entry screen of BiciSegura. It has a banner with the app slogan (“Your city. Your bike. Your safe route.”) and two tabs, Sign in and Register. The user types an email and a password, or taps “Continue with Google”. There is also a “Forgot your password?” link and a note that accepting the terms is required. Without an active session the user cannot reach the rest of the app. After signing in, the system opens the cyclist or administrator view according to the user’s role.

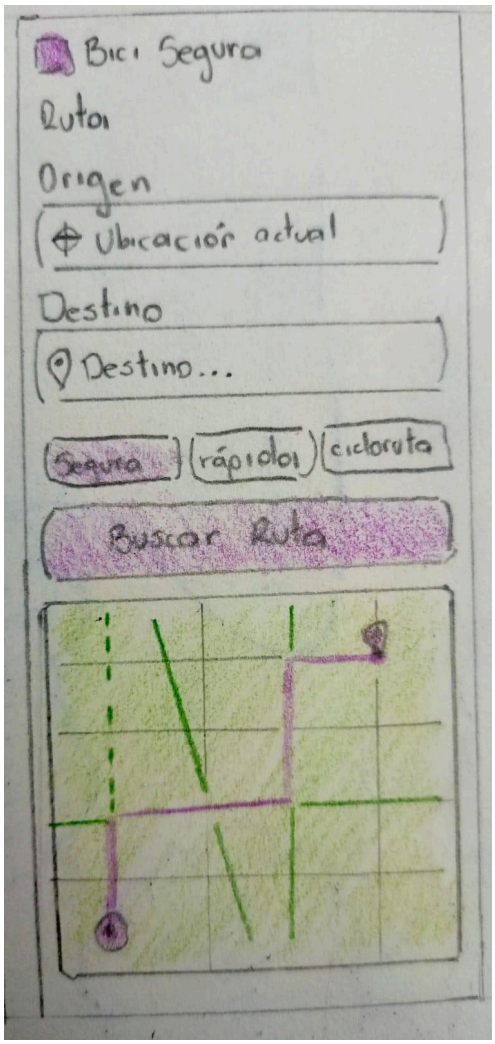




Plan a Route

This is the main screen of the cyclist flow. The user types an origin (the current location is the default) and a destination, and chooses a route preference: Safest, Fastest or Bike lanes. Two checkboxes let them avoid high-speed roads and unpaved streets. Pressing “Search routes” calculates the options. Below the form is a map preview of the area with the bike lanes layer, and a safety tip at the bottom. A navigation bar gives access to the other sections (Routes, Incidents, Report, Profile).

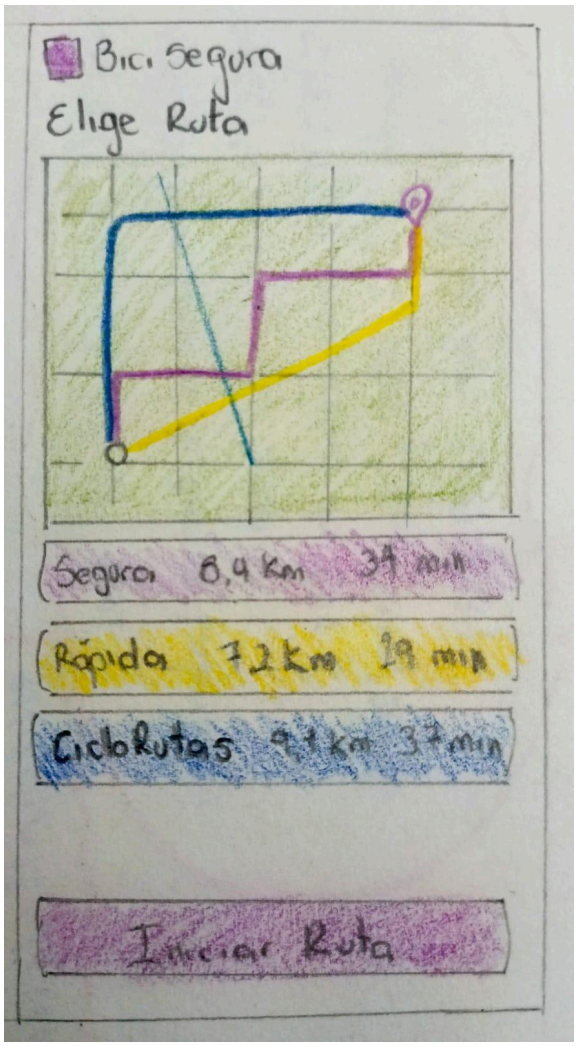




Compare Alternatives

After the search, this screen shows a map with the routes drawn in different colors (A, B and C) and a list of the routes found, ordered by safety. Each card shows the distance, the estimated time, the percentage of the route on bike lanes and a safety score with a label (Safe, Medium). Route A is marked as the safest and recommended, B as the fastest and C as the one with the most bike lanes. The cyclist taps a card to select it and presses “Start route A”, or goes back with “Change origin or destination”.



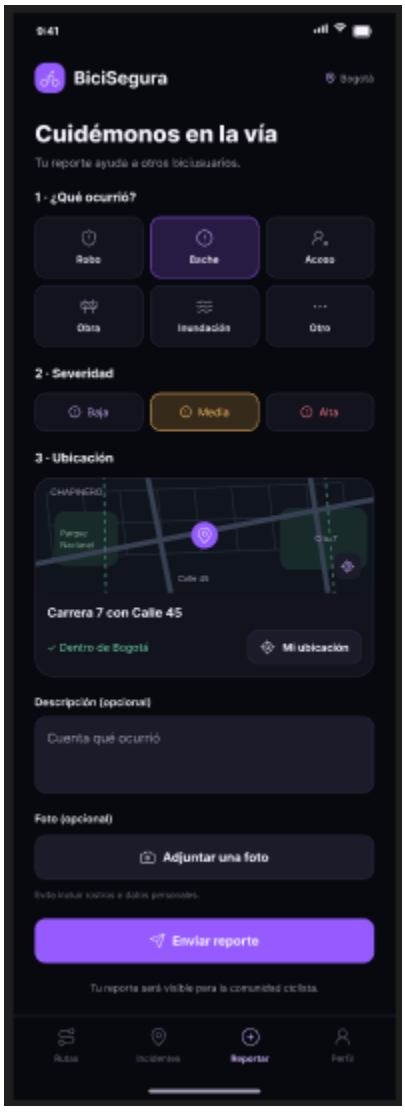


Report an Incident

This is a three-step form to report a hazard.

- The user chooses the type (robbery, pothole, harassment, roadwork, flooding or other).
- They set the severity (low, medium or high).
- They confirm the location on a small map, using their current position or by moving the pin. The system checks that the location is inside Bogotá.

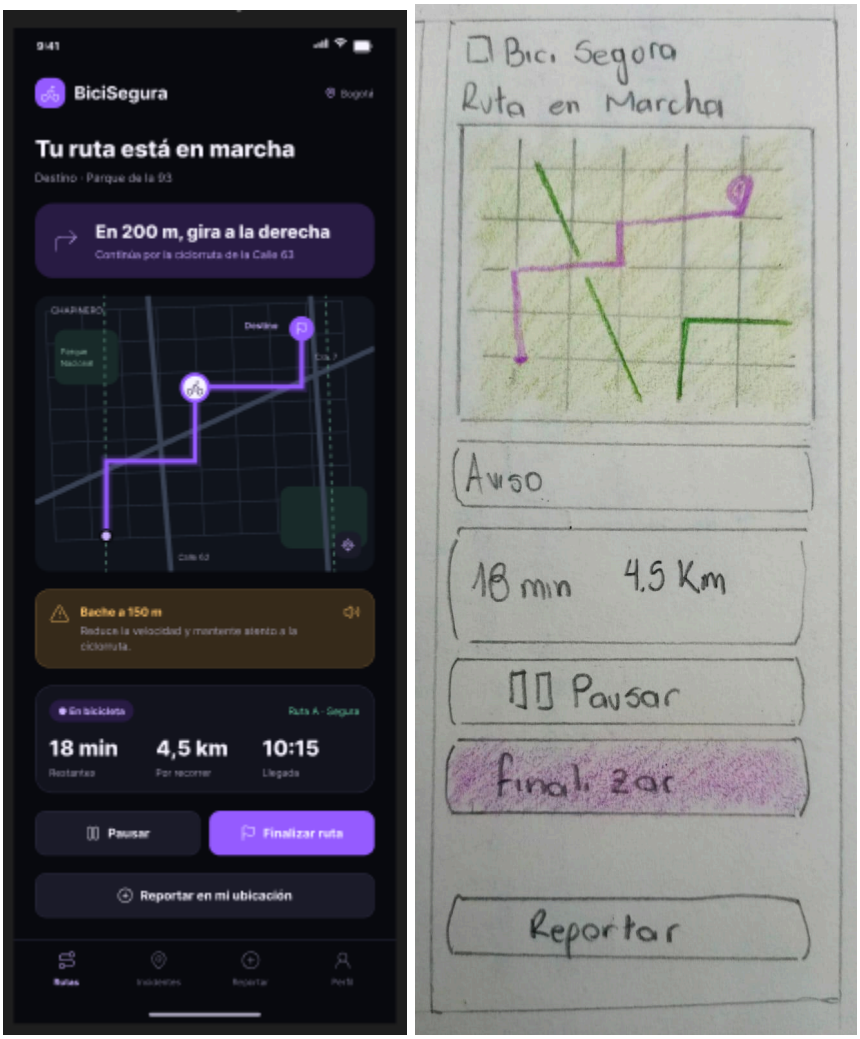
There is also an optional description, an optional photo and a “Send report” button. The report becomes visible to the rest of the cyclist community.





Active Ride & Alerts

This is the live navigation screen. At the top, a banner gives the next instruction (“In 200 m, turn right”). The map shows the route and the cyclist’s position, and a yellow alert card warns about a hazard ahead (“Pothole in 150 m”). A panel at the bottom shows the remaining time, the distance and the estimated arrival time. The cyclist can Pause or Finish route, and can also report an incident from their current location without leaving the ride.



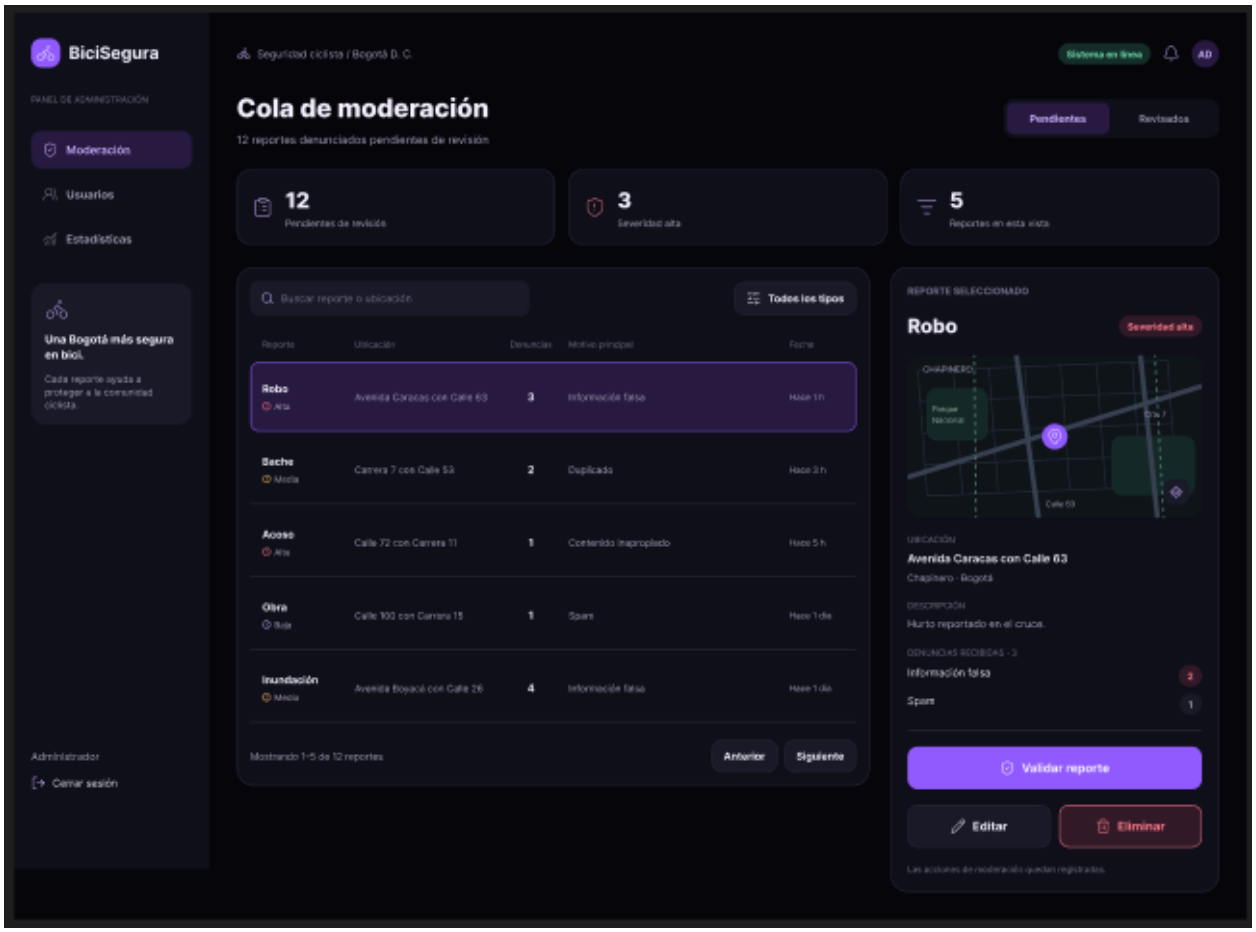
Incidents, Filters & Detail

This screen is a community safety map with three tabs: Incidents, Unsafe zones and Bike lanes. In the first tab the incidents appear as colored markers (high, medium, low severity), and filters let the user narrow them by type and by time (for example, last 7 days). Unsafe zones are drawn as highlighted areas with a risk index. Tapping an incident opens a detail card (for example, a robbery on a specific street) with the severity, the time, the description and the number of confirmations. Two buttons let the user confirm whether it is “Still happening” or “Already solved”, and a “Report an incident” button leads to the next screen.



Moderation Queue (Admin)

This is the administrator's desktop panel, with a side menu for Moderation, Users and Statistics. At the top, counters show the reports pending review, the high-severity ones and the ones in the current view. The main table lists the reported reports with their type, location, number of complaints, main reason (false information, duplicate, spam, inappropriate content) and date, and it has a search box and a type filter. When the admin selects a row, a side panel shows the map, the address, the description and the complaints received. From there the admin can Validate the report, Edit it or Delete it, and the system records each action.




users
bigint user_id PK
varchar full_name
varchar email UK
varchar password_hash
varchar role
varchar status
timestamp privacy_consent_at
smallint failed_login_attempts
timestamp locked_until
timestamp created_at
geocoding_cache
varchar query_text PK
geometry location
timestamp cached_at
has
learns
request
author
has
perform
user_profiles
bigint user_id PK,FK
varchar bicycle_type
varchar fitness_level
reputation_accounts
bigint user_id PK,FK
integer points
varchar level
password_reset_tokens
bigint token_id PK
bigint user_id FK
varchar token_hash UK
timestamp expires_at
timestamp used_at
incident_reports
bigint report_id PK
bigint author_id FK
varchar type
varchar severity
text description
geometry location
varchar address
varchar state
integer confirmation_count
integer solved_count
numeric current_weight
timestamp created_at
timestamp updated_at
travel_preferences
bigint user_id PK,FK
varchar priority
boolean avoid_high_speed_roads
boolean avoid_unpaved_streets
search_history
bigint search_id PK
bigint user_id FK
varchar origin_text
geometry origin_point
varchar destination_text
geometry destination_point
varchar priority
timestamp searched_at
cycle_routes
bigint cycle_route_id PK
varchar name
geometry geometry
varchar source_attribution
owns
raises
execute
vote
receives
receives
target
produce
covers
report_votes
bigint vote_id PK
bigint report_id FK
bigint user_id FK
varchar vote_type
timestamp created_at
report_flags
bigint flag_id PK
bigint report_id FK
bigint flagged_by FK
varchar reason
varchar status
bigint reviewed_by FK
timestamp created_at
timestamp reviewed_at
moderation_actions
bigint action_id PK
bigint admin_id FK
bigint target_user_id FK
bigint target_report_id FK
varchar action_type
text notes
timestamp created_at
route_alternatives
bigint alternative_id PK
bigint search_id FK
smallint rank_position
geometry geometry
numeric distance_m
numeric adjusted_time_min
numeric cycle_route_pct
numeric safety_score
varchar safety_color
boolean was_selected
road_segments
bigint segment_id PK
bigint osm_way_id
geometry geometry
bigint cycle_route_id FK
numeric safety_index
timestamp last_calculated_at
followed_in
composed_of
used_in
navigation_sessions
bigint session_id PK
bigint user_id FK
bigint alternative_id FK
varchar status
timestamp started_at
timestamp ended_at
alternative_segments
bigint alternative_id PK,FK
bigint segment_id PK,FK
smallint segment_order